The Dot Product as Projection, and Duality

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Where this fits: Pipeline map step 0 (Foundations). See the Course map.

0 Foundations	1 Frame	2 Get data	3 Understand	4 Clean	5 Features	6 Reduce
7 Split	8 Model	9 Evaluate	10 Tune	11 Deploy	12 Test	13 Monitor

- **Builds on:** Dot product.

1. Overview

Key point: The dot product $\mathbf{v} \cdot \mathbf{w}$ is the signed length of $\mathbf{w}$'s shadow on the line of $\mathbf{v}$ (negative when the shadow points away from $\mathbf{v}$) times the length of $\mathbf{v}$; and every linear function that turns a vector into a number is a dot product with one particular vector.

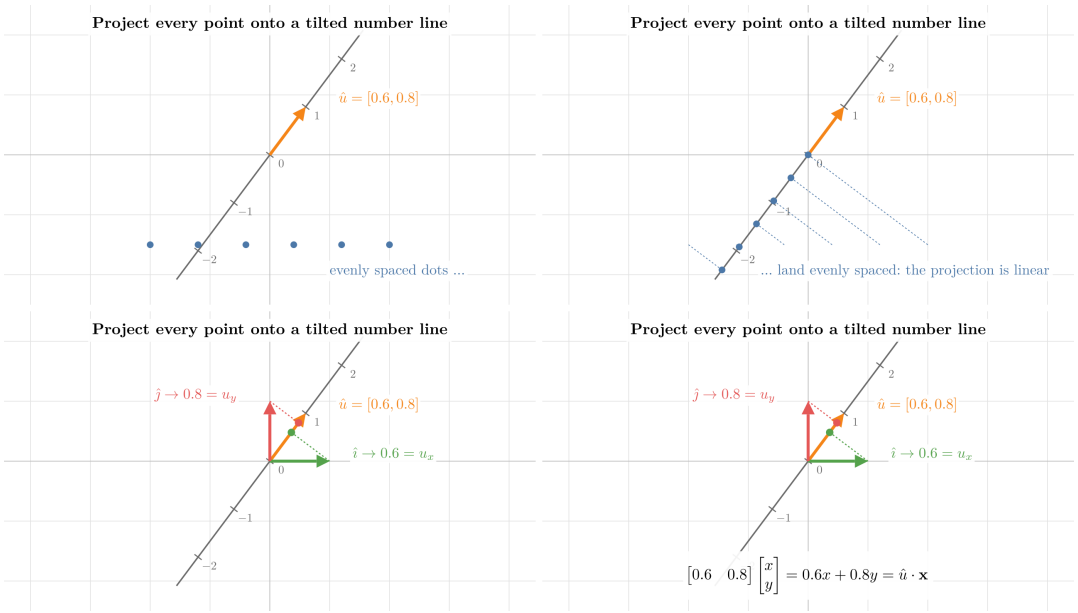


Figure 1: Projecting the plane onto a tilted number line is a linear transformation whose 1×2 matrix holds the coordinates of $\hat{u}$; applying it is a dot product with $\hat{u}$

Computing the dot product taught the rule "multiply matching components and add" (a component is one entry of the list), its two laws and its geometric form $\|a\|\|b\| \cos\theta$. This Note adds two views that those sections do not cover:

- **Projection.** The dot product as "shadow length times length", and why that view does not depend on which vector casts the shadow (Sections 2 and 3).
- **Duality.** Why the component recipe has anything to do with **projection** (G-1583): linear transformations from vectors to numbers are exactly dot products (Sections 4 to 6, and Figure 1).

The Note uses linear transformations (functions that keep grid lines parallel and evenly spaced) and their matrices from the matrix of a transformation. The symbols $\hat{\imath} = [1, 0]$

and $\hat{\jmath} = [0, 1]$ are the two arrows of length 1 along the x-axis and y-axis; where a linear transformation sends them decides where it sends everything (two vectors decide everything).

2. The dot product as a projection

> **Key point:** Project **w** onto the line through **v**, then multiply the length of that projection by the length of **v**; the result is negative when the projection points away from **v**.

Recall the numeric rule: pair up the coordinates, multiply each pair, add. For example:

$$[1, 2] \cdot [3, 4] = 1 \times 3 + 2 \times 4$$

$$[1, 2] \cdot [3, 4] = 11$$

The rule has a picture.

PCA (principal component analysis) uses the **projection** (G-1583) (shadow) of a point onto a line, and projecting one point showed that its length is $u^\mathsf{T}x$ when u is a unit vector (a vector of length 1). Here the line's vector need not have length 1, and that gives a full picture of the dot product.

1. **In words:** drop **w** straight onto the line through the origin and **v**. Measure the length of this shadow, with a minus sign if it points opposite to **v**. Multiply by the length of **v**.
2. **Formula:**

   $$\mathbf{v} \cdot \mathbf{w} = (\text{signed shadow length}) \times \|\mathbf{v}\|$$

   Here "signed shadow length" is the signed length of the projection of **w** onto **v**.
3. **Example:** for $\mathbf{v} = [3, 1]$ and $\mathbf{w} = [1, 2]$ (Figure 2, first stop), the shadow of **w** ends at $[1.5, 0.5]$, half of **v**. We can check that this is the foot of the drop: the leftover piece must be at 90° to **v**.

   $$[1, 2] - [1.5, 0.5] = [-0.5, 1.5]$$

   $$(-0.5)(3) + (1.5)(1) = -1.5 + 1.5 = 0$$

   The shadow's length and the length of **v** are:

   $$\sqrt{1.5^2 + 0.5^2} \approx 1.58$$

   $$\|\mathbf{v}\| = \sqrt{10} \approx 3.16$$

The dot product, first as shadow times length, then by components:

$$\mathbf{v} \cdot \mathbf{w} = 1.58 \times 3.16 = 5$$

$$3 \times 1 + 1 \times 2 = 5$$

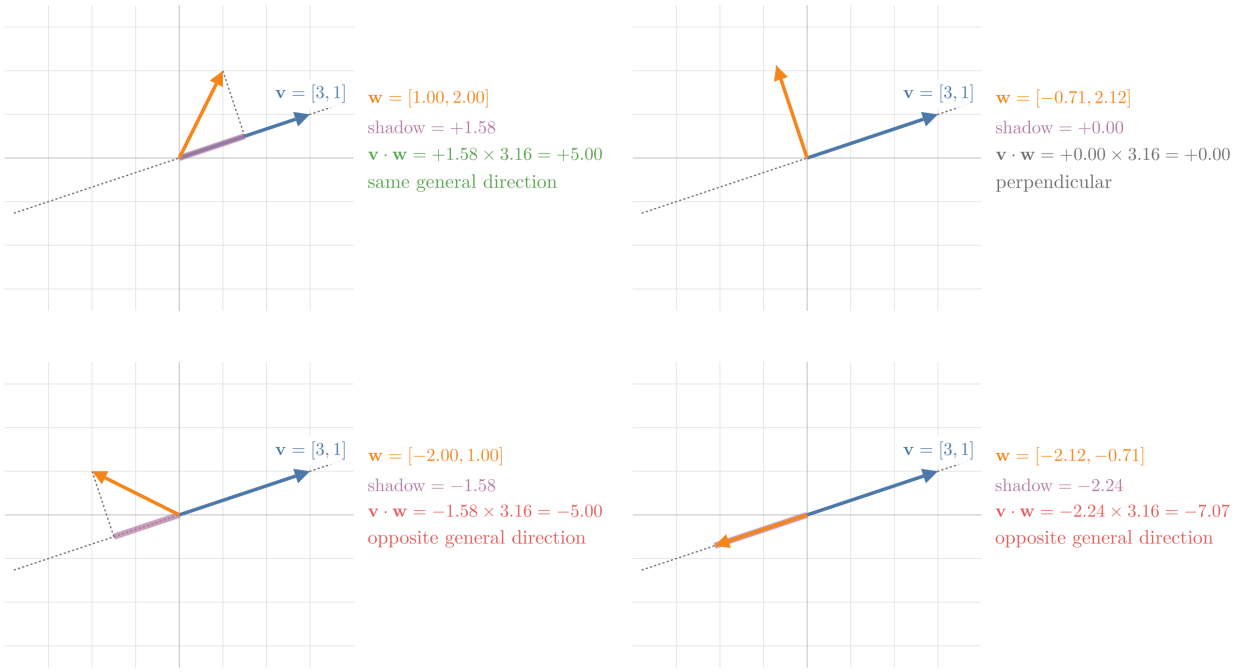


Figure 2: w turns once around the origin while v = [3, 1] stays fixed; the purple bar is the shadow of w on the line of v, and the readout multiplies its signed length by the length of v. Picture after 3Blue1Brown, "Dot products and duality"

In Figure 2, $\mathbf{w}$ turns once around the origin. Watch the purple shadow: it shrinks to nothing when $\mathbf{w}$ is perpendicular to $\mathbf{v}$, then grows on the other side, and the dot product changes sign with it. The animation stops at cases like the three below (the readout names each one: same general direction, perpendicular, opposite general direction). These are the signs of perpendicular vectors have dot product 0, now read as shadows:

- **Same general direction:** the shadow points along $\mathbf{v}$, so the dot product is positive.
- **Perpendicular:** the shadow is just the origin, the zero vector, so the dot product is 0.
- **Opposite general direction:** the shadow points away from $\mathbf{v}$, so the dot product is negative. With $\mathbf{w} = [-2, 1]$, the shadow ends at $[-1.5, -0.5]$ and:

$$\mathbf{v} \cdot \mathbf{w} = -6 + 1 = -5$$

Extra: The projection view is the formula $\|a\|\|b\|\cos\theta$ of the second formula, regrouped. In a right triangle, $\|\mathbf{w}\|\cos\theta$ is the side along $\mathbf{v}$: exactly the signed length of the shadow.

3. Why the order does not matter

Key point: Projecting $\mathbf{w}$ onto $\mathbf{v}$ and projecting $\mathbf{v}$ onto $\mathbf{w}$ look like different processes, but they give the same number.

The projection view treats the two vectors very differently: one casts the shadow, the other receives it. We could instead project $\mathbf{v}$ onto $\mathbf{w}$ and multiply by $\|\mathbf{w}\|$. The answer is the same, because $\mathbf{v}\cdot\mathbf{w} = \mathbf{w}\cdot\mathbf{v}$. Two steps explain why.

- **Equal lengths: symmetry.** If $\mathbf{v}$ and $\mathbf{w}$ have the same length, the picture is symmetric. Projecting $\mathbf{w}$ onto $\mathbf{v}$ is the mirror image of projecting $\mathbf{v}$ onto $\mathbf{w}$, across the line halfway between them, so both products are equal.
- **Scaling one vector.** Now stretch $\mathbf{v}$ to $2\mathbf{v}$, which breaks the symmetry. If $\mathbf{w}$ is projected onto $2\mathbf{v}$, the shadow is unchanged and the length we multiply by doubles. If $2\mathbf{v}$ is projected onto $\mathbf{w}$, the shadow doubles and $\|\mathbf{w}\|$ is unchanged. Either way the dot product doubles, so the two views still agree.

Any two vectors can be reached this way from two vectors of equal length, so the two views always agree.

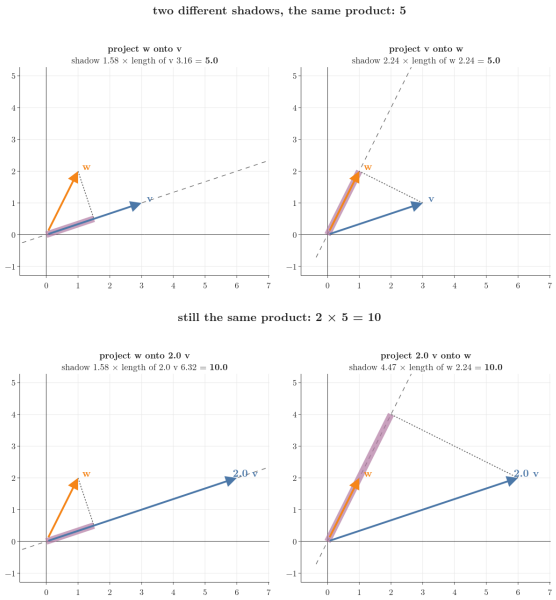


Figure 3: Both views on Section 2's vectors: the shadows differ, the products agree (5), and after stretching v to 2v both give 10

In Figure 3, watch the two products as $\mathbf{v}$ stretches: on the left only the length grows, on the right only the shadow grows, and the two numbers stay equal.

4. Linear transformations from vectors to numbers

> **Key point:** A linear transformation from 2D to the number line is fixed by the two numbers where $\hat{\imath}$ and $\hat{\jmath}$ land, so its matrix is 1×2; applying it looks exactly like a dot product.

Why should "multiply matching components and add" have anything to do with **projection** (G-1583)? The answer, called duality, needs one more kind of linear transformation: one whose output is a single number, a point on the number line.

4.1 The visual test

> **Key point:** A transformation to the number line is linear when evenly spaced dots on any line land evenly spaced.

For a transformation to the number line, "grid lines stay parallel and evenly spaced" becomes: take any line of evenly spaced dots in the plane; after the transformation, the dots must still be evenly spaced on the number line. If some line of evenly spaced dots lands unevenly, the transformation is not linear.

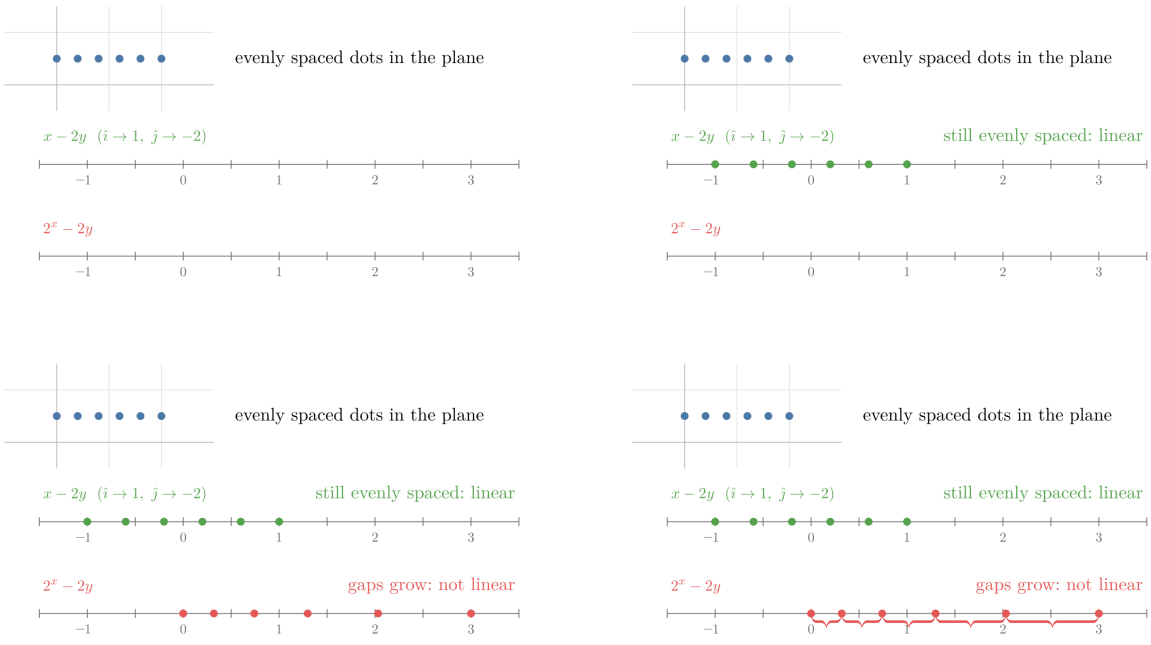


Figure 4: The visual test: six evenly spaced dots sent to the number line by x - 2y (linear) and by 2^x - 2y (not linear). Test after 3Blue1Brown, "Dot products and duality"

Figure 4 runs the test on six dots, spaced 0.4 apart on the line $y = 0.5$:

- $x - 2y$ sends them to $-1, -0.6, -0.2, 0.2, 0.6, 1$: still 0.4 apart, so it passes.
- $2^x - 2y$ sends them to $0, 0.32, 0.74, 1.30, 2.03, 3$: the gaps grow, so it is not linear.

4.2 The 1×2 matrix

> **Key point:** Each basis vector lands on a number, so each column of the matrix has one entry.

As always, the transformation is fixed by where $\hat{\imath}$ and $\hat{\jmath}$ land; now each lands on a single number. Written as columns, the two numbers make a 1×2 matrix.

1. **In words:** split the vector into its coordinates times $\hat{\imath}$ and $\hat{\jmath}$, and replace $\hat{\imath}$ and $\hat{\jmath}$ by the numbers they land on.
2. **Formula:** if $\hat{\imath}$ lands on a and $\hat{\jmath}$ on b,

$$\begin{bmatrix} a & b \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = a\,x + b\,y$$

3. **Example:** if $\hat{\imath}$ lands on 1 and $\hat{\jmath}$ on -2, the vector $[4, 3]$, which is 4 steps along $\hat{\imath}$ and 3 along $\hat{\jmath}$, lands on:

$$\begin{bmatrix} 1 & -2 \end{bmatrix} \begin{bmatrix} 4 \\ 3 \end{bmatrix}$$

$$= 4 \times 1 + 3 \times (-2)$$

$$= -2$$

That computation is the dot product $[1, -2] \cdot [4, 3]$. A 1×2 matrix looks just like a 2D vector tipped on its side, and multiplying it by a vector is the same arithmetic as the dot product with the upright vector. The 1×2 matrix product is the row-times-column form $a^{\mathsf{T}}b$ of the dot product as a matrix product, now read as a transformation.

5. Projection onto a tilted number line

> **Key point:** Projecting the plane onto a number line along the unit vector $\hat{u}$ is linear, and by symmetry its matrix is $[u_x \;\; u_y]$; so projecting is the dot product with $\hat{u}$.

Now forget that we already know the dot product relates to **projection** (G-1583), and build it from scratch (Figure 1).

5.1 Projecting is linear

> **Key point:** Dropping every point straight onto a tilted line through the origin defines a linear transformation to numbers.

Place a copy of the number line diagonally in the plane, with 0 at the origin. Let $\hat{u}$ be the 2D unit vector whose tip sits at the number 1 on this line; in Figure 1, $\hat{u} = [0.6, 0.8]$. Projecting each point of the plane straight onto the line gives a number: its position on the line. **Projection** (G-1583) is a function from 2D vectors to numbers, and the function passes the visual test: evenly spaced dots land evenly spaced (Figure 1, the stage where the blue dots drop onto the line). So projection is linear, and it has a 1×2 matrix.

5.2 Where $\hat{\imath}$ and $\hat{\jmath}$ land

> **Key point:** $\hat{\imath}$ lands on u_x and $\hat{\jmath}$ on u_y, because projecting $\hat{\imath}$ onto $\hat{u}$ mirrors projecting $\hat{u}$ onto the x-axis.

To find the matrix, we ask where $\hat{\imath}$ and $\hat{\jmath}$ land (Figure 1, the stage with the green and red arrows).

- **$\hat{\imath}$:** $\hat{\imath}$ and $\hat{u}$ are both unit vectors, so projecting $\hat{\imath}$ onto the line of $\hat{u}$ is the mirror image of projecting $\hat{u}$ onto the x-axis. The second is simply the x-coordinate of $\hat{u}$. So $\hat{\imath}$ lands on $u_x = 0.6$.
- **$\hat{\jmath}$:** by the same symmetry, with the y-axis, $\hat{\jmath}$ lands on $u_y = 0.8$.

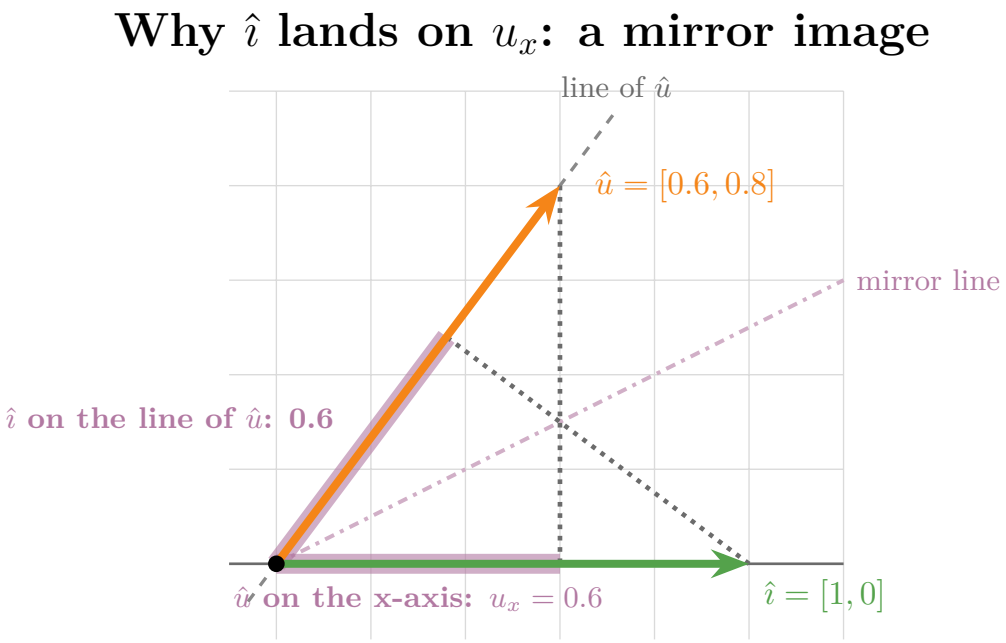


Figure 5: Projecting the unit vector i onto the line of u (0.6) mirrors projecting u onto the x-axis (u_x = 0.6)

In Figure 5, fold the picture along the mirror line: $\hat{\imath}$ swaps with $\hat{u}$, and the two purple shadows swap too, so both have length 0.6.

So the matrix of the **projection** (G-1583) is $[u_x \;\; u_y] = [0.6 \;\; 0.8]$: the coordinates of $\hat{u}$, tipped on their side. Applying it to any vector is the dot product with $\hat{u}$.

1. **In words:** the position of a vector's shadow on the line of a unit vector $\hat{u}$ is the dot product with $\hat{u}$.
2. **Formula:**

$$\begin{bmatrix} u_x & u_y \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

$$= u_x\, x + u_y\, y$$

$$= \hat{u} \cdot \mathbf{x}$$

3. **Example:** for $\hat{u} = [0.6, 0.8]$ and $\mathbf{x} = [2, 1]$:

$$\hat{u} \cdot \mathbf{x} = 0.6 \times 2 + 0.8 \times 1$$

$$\hat{u} \cdot \mathbf{x} = 2.0$$

So $\mathbf{x}$ lands on the number 2.0 of the tilted line.

5.3 Non-unit vectors

> **Key point:** Scaling $\hat{u}$ by 3 multiplies every output by 3: the dot product with a non-unit vector is "project, then multiply by its length".

Scale $\hat{u}$ by 3, to $[1.8, 2.4]$. Its 1×2 matrix sends $\hat{\imath}$ and $\hat{\jmath}$ to 3 times their old numbers, so by linearity every vector lands on 3 times its old number. For $\mathbf{x} = [2, 1]$:

$$1.8 \times 2 + 2.4 \times 1 = 6.0$$

$$6.0 = 3 \times 2.0$$

So the dot product with a vector of any length is: project onto its line, then multiply by its length. This result is the projection view of Section 2, now derived instead of assumed.

> **Python:** The three numbers of this section.

```python
import numpy as np

u = np.array([0.6, 0.8])            # unit vector
```

```
x = np.array([2, 1])
u @ x                          # 2.0: position of the shadow
(3 * u) @ x                    # 6.0: three times as much
np.array([[1, -2]]) @ [4, 3]   # array([-2]): a 1 x 2 matrix
```

6. Duality

> **Key point:** Every linear transformation from vectors to numbers is a dot product with one unique vector, and every vector defines such a transformation.

Look back at what happened. We defined a linear transformation from the plane to the number line by pure geometry, **projection** (G-1583), with no dot product in sight. Because it is linear, it has a 1×2 matrix. And because a 1×2 matrix times a vector is the same arithmetic as a dot product, the transformation is a dot product with some vector, here $\hat{u}$.

Figure 6 shows the same thing for a transformation defined with no vector at all, only by where $\hat{\imath}$ and $\hat{\jmath}$ land (1 and -2). Watch the grid points squish onto the number line, $[4, 3]$ landing on -2; then the 1×2 matrix is tipped upright into the vector $\mathbf{v} = [1, -2]$, and projecting $[4, 3]$ onto its line and multiplying by its length gives the same -2. The picture follows Sanderson's *Essence of Linear Algebra*, chapter 9 (3Blue1Brown).

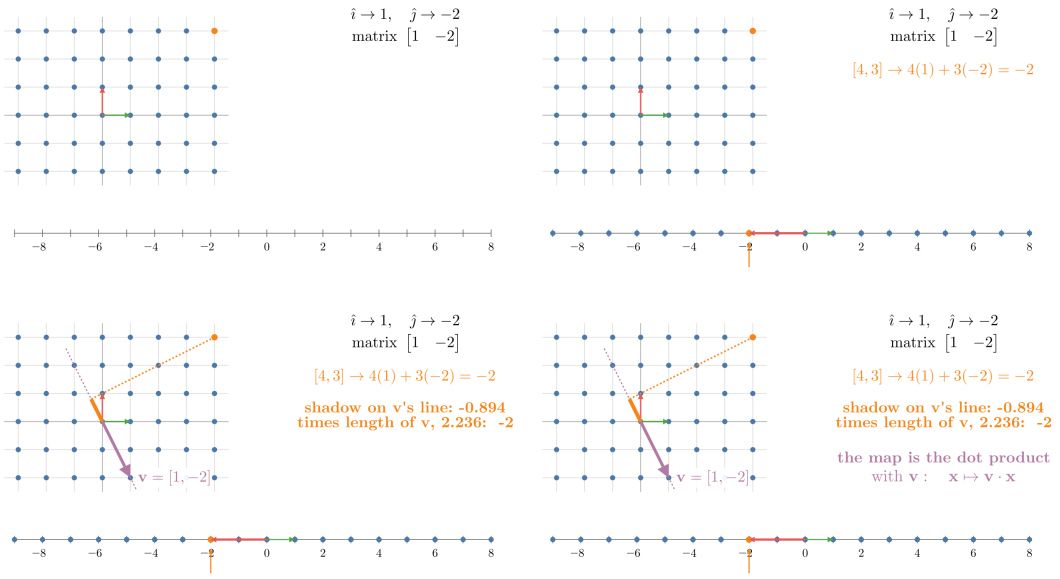


Figure 6: Duality: a linear map to numbers, fixed by where the basis vectors land, sends every grid point to $x_1 - 2x_2$; the same numbers read as the vector $[1, -2]$ give the same answer as a projection

The same holds for every linear transformation whose output is a number, however it was defined: there is exactly one vector $\mathbf{v}$ such that applying the transformation is the same as taking the dot product with $\mathbf{v}$. The vector is the transformation's matrix tipped upright.

This natural but surprising correspondence between two kinds of objects is an example of **duality** (G-644). For linear algebra:

- the dual of a vector is the linear transformation to numbers that it encodes, $\mathbf{x} \mapsto \mathbf{v} \cdot \mathbf{x}$;
- the dual of a linear transformation from a space to numbers is a vector in that space.

On the surface, the dot product is a tool for projections and for testing whether two vectors point the same way. Underneath, taking a dot product with $\mathbf{v}$ turns $\mathbf{v}$ into a transformation. A vector can be thought of as an arrow, or as a compact way of writing down a function that sends space to the number line.

7. Where ML uses duality

> **Key point:** A linear model's weight vector is the dual of its scoring function; each neuron is one such dual vector, and a weight matrix stacks them as rows.

7.1 The score of a linear model

> **Key point:** $w^\mathsf{T}x + w_0$ projects x onto the direction of w, scales by $\|w\|$ and shifts.

Linear models score a feature vector x (one value per **feature** (G-772), an input variable) with $w^\mathsf{T}x + w_0$, the left side of the hyperplane equation (a hyperplane is a flat surface, such as a line in 2D; see the vector form). The part $w^\mathsf{T}x$ is a linear transformation from feature vectors to numbers, and w is its dual vector. Read as a **projection** (G-1583): the score measures how far x reaches along the direction of w, times $\|w\|$, plus the shift w_0. Points with equal shadows on the line of w get equal scores; the hyperplane is the set of points whose score is 0.

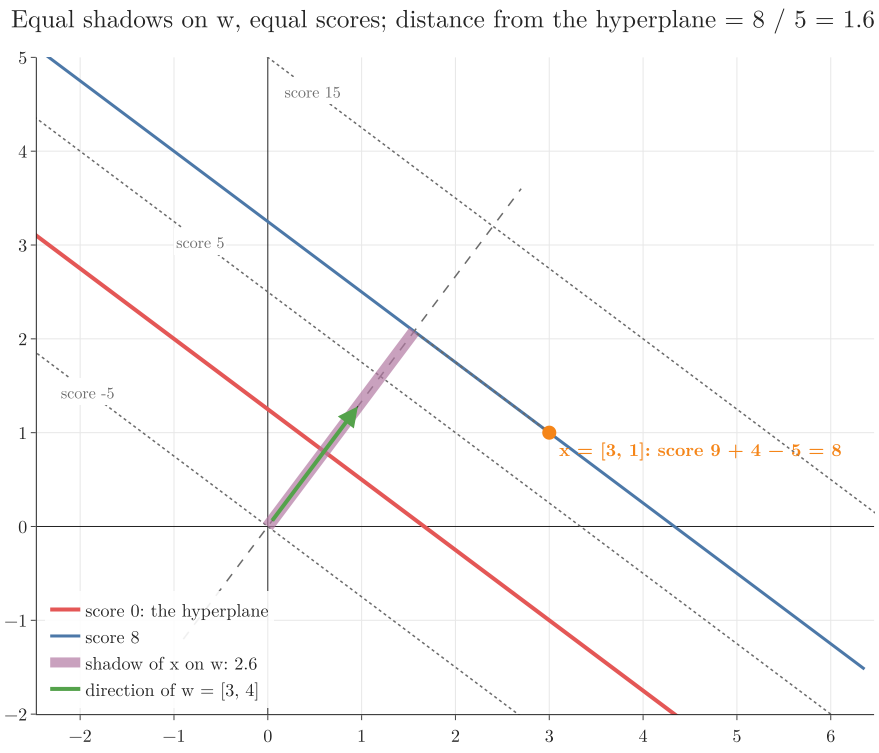


Figure 7: The score of a linear model with w = [3, 4] and shift -5: every line perpendicular to w is a line of equal score

In Figure 7, slide $x = [3, 1]$ along the blue line: its shadow on w stays at 2.6, so its score stays at 8. The Extra below computes its distance from the red hyperplane.

> **Extra:** Dividing the score by $\|w\|$ removes the "times the length" and leaves the signed distance of x from the hyperplane:
>
> $$(w^\mathsf{T} x + w_0)/\|w\|$$
>
> For $w = [3, 4]$, $w_0 = -5$ and $x = [3, 1]$, the score and the distance are:
>
> $$9 + 4 - 5 = 8$$
>
> $$8/5 = 1.6$$
>
> SVM's margin is built on this signed distance (see the margin in terms of w).

7.2 A matrix as stacked dual vectors

> **Key point:** Each row of a weight matrix is one linear function to numbers; $W\mathbf{x}$ computes all of them at once.

A $1 \times n$ matrix is one linear function from n **features** (G-772) to a number: one neuron's

weighted sum, or one principal component's score. A matrix with k rows stacks k such functions, and each entry of $W\mathbf{x}$ is the dot product of one row with $\mathbf{x}$. Stacking rows gives the row-by-row reading of matrix multiplication, next to the column-by-column reading of computing a product column by column.

- **A layer of neurons:** row j of the weight matrix is the dual vector of neuron j; the neuron's output before the activation is the **projection** (G-1583) of the input onto that row, scaled.
- **PCA:** each principal component is a unit vector u, and a point's score on it is $u^{\mathsf{T}}x$, the position of its shadow on the line of u (see the projection step).

> **Extra:** Recommender systems often give every user and every item a learned vector, an **embedding** (G-677), and predict a rating as their dot product (Koren et al. 2009). By **duality** (G-644), a user's vector is a linear scoring function over items: it projects each item's vector onto the user's taste direction.

8. Summary

View	What $\mathbf{v} \cdot \mathbf{w}$ is	Example	Why it matters
Components	multiply matching components, add	$3 \times 1 + 1 \times 2 = 5$	the quick way to compute it
Projection	signed shadow length $\times \|\mathbf{v}\|$	$1.58 \times 3.16 = 5$	gives it a meaning: how far one vector reaches along the other
Transformation	1×2 matrix $\mathbf{v}^\mathsf{T}$ applied to $\mathbf{w}$	$[1 \quad -2]\,[4, 3] = -2$	shows the component rule is a linear map to numbers
Duality	every linear map to numbers is a dot product with one vector	projection onto $\hat{u}$ = dot with $[0.6, 0.8]$	explains why the component rule and the shadow agree

- The dot product is positive, zero or negative as the shadow points along, vanishes, or points against $\mathbf{v}$, so the sign alone says whether two vectors point the same general way.
- Which vector casts the shadow does not matter, because equal-length vectors are mirror images, and stretching one vector stretches either the shadow or the length by the same factor.
- **Projection** (G-1583) onto the line of a unit vector $\hat{u}$ is linear, and its matrix is $\hat{u}$ tipped on its side, because by mirror symmetry $\hat{\imath}$ lands on u_x and $\hat{\jmath}$ on u_y.
- A linear model's weights, a neuron's weights and a principal component are all dual vectors: linear functions to numbers written as arrows, so a model's score can be read as a shadow on the line of its weight vector.
- So the component recipe and the shadow picture are one thing, because every linear function from vectors to numbers is a dot product with one vector.

9. Sources

Built from

- Sanderson, G. (3Blue1Brown), "Dot products and duality | Chapter 9, Essence of linear algebra", 2016, 3blue1brown.com/lessons/dot-products, https://www.youtube.com/watch?v=LyGKyc

Other references

- Koren, Y., Bell, R. and Volinsky, C. (2009). "Matrix Factorization Techniques for Recommender Systems". *IEEE Computer* 42(8).

10. Key terms

Terms taught in this Note come first; linked terms are recaps, taught in the Note the link opens.

Term	Meaning
Duality (G-644)	The one-to-one match between vectors and linear maps that turn a vector into a number: every such map is a dot product with exactly one vector. So a vector can be read as a scoring function, and a scoring function as a vector.
Dual vector (G-643)	The one vector whose dot product does the same job as a given linear transformation from vectors to a number; for example, a linear model's weight vector is the dual vector of its scoring function.
Linear transformation to the number line (G-1096)	A linear transformation that sends each vector to a single number, keeping evenly spaced dots evenly spaced; its matrix is one row ($1 \times n$), so applying it is the same as a dot product with that row.
Projection view of the dot product (G-1582)	Reading the dot product as a shadow: $\mathbf{v} \cdot \mathbf{w}$ is the signed length of $\mathbf{w}$'s shadow (projection) on the line of $\mathbf{v}$ times the length $\|\mathbf{v}\|$, negative when the shadow points away from $\mathbf{v}$.
Projection (G-1583)	Dropping a point or vector straight onto an axis, a line or another vector's direction, like casting a shadow.
Feature (G-772)	One piece of information about each example that a model uses (e.g. a student's CGPA).
Embedding (G-677)	A learned vector representing a user, item or word, compared with others by dot products.